

P1 as an overload bound

$\text{ind}_-(A_0) \leq 1$ is a crossing budget of the μ -CD chain on Y , not a Cauchy- π identity

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Abstract

The reviewer reframes P1 as $\text{ind}_-(A_0) \leq 1$ on the selected sequence (vacuous windows are the first case, not an error). The object $A_0 = R_{N-3} - \frac{1}{2}I$ lives three ranks before the end of the window. The last exact dictionary is $R = C(I+C)^{-1}$ (r407), so $\text{ind}_-(A_0) = n_C(N_w - 3)$ with $n_C(n) = \#\{\lambda(C(n)) < 1\}$ and $C(n)$ the μ -only dual CD Gram on the holes Y . The nested OP matrix gives the constructor identity $C(n+1) - C(n) = vv^T$ (rank-one Loewner); hence n_C is antitone and drops by at most one (**rankOne_inertia_antitone**). The unique interpolant at depth $d = |Y|$ still has $n_C \gg 1$ (flagship 43). Extra OP ranks lift the excess down to 0 or 1 on every selected window, and get stuck at 19–21 on permute / jitter / scramble. A universal Gantmacher–Krein cap $\text{ind}_-(W_Y - S_0) \leq 1$ for the r431 Cauchy- π residual is refuted already on 2×2 and 3×3 toys over $\mathbb{Q}$. No RH claim.

Status. Lemmas 1–2 are proved (finite identities). Propositions 1–3 are finite machine checks. Ausgang DEPTH_CHAIN_EXACT / GK_UNIVERSAL_REFUTED / SAVE_RATE_CENSUS. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The nested CD chain

Let Y be the holes of a window, $u^\vee$ the r356 dual weights, and B_n the n -column μ -OP matrix on Y dressed by $\sqrt{u^\vee|_Y}$ (r407 chain_C). Write $C(n) = B_n B_n^T$.

Lemma 1 (nested rank-one Loewner). *If $B_n = [B_{n-1} \mid v]$ then $C(n) - C(n-1) = vv^T \succeq 0$ of rank at most one. Consequently $n_C(n)$ is nonincreasing and $n_C(n-1) - 1 \leq n_C(n) \leq n_C(n-1)$ (Cauchy interlacing; Lean **rankOne_inertia_antitone**). At working depth, $n_C(N_w - 3) = \text{ind}_-(A_0)$ (r407/r412 graph-resolvent dictionary).*

The identity is the constructor, not a fit. On flagship MAIN ($k_z = 9$, $|Y| = 104$, $N_w - 3 = 181$) every sampled consecutive update has numerical rank 1 (Loewner floor -10^{-13}).

Lemma 2 (alternating Δ). *For ordered holes $y_1 < \dots < y_q$, the barycentric weights $\pi'_Y(y_j) = \prod_{k \neq j} (y_j - y_k)$ alternate in sign, $\text{sign}(\pi'_Y(y_j)) = (-1)^{q-1-j}$.*

This is the sign bookkeeping of the r431 Cauchy- π formula $S_0 = \Delta^{-1} C \Pi W_X \Pi C^T \Delta^{-1}$. It does *not* cap $\text{ind}_-(W_Y - S_0)$.

2 Gantmacher–Krein at interpolating depth

At $d_0 = |Y|$ the interpolant is unique (kernel dimension 0) and S_0 is the Cauchy- π kernel of the $\pi_Y(X)^2$ -dressed X -measure (r431, exact over $\mathbb{Q}$). Cauchy kernels of positive masses are

totally positive; Δ is alternating. The candidate “ W_Y minus a sign-regular dressing has at most one negative square” is *false* as a universal identity.

Proposition 1 (no universal n_- cap at $k_{\dim} = 0$). *On the r409 five-atom toy at $d_0 = |Y| = 2$, $D_0 = W_Y - S_0$ has inertia $(1, 1, 0)$ and $\det D_0 = -3740/243 < 0$ (one negative square); S_0 and the undressed Pick P_X are positive definite. Raising the depth by one lifts D_0 to inertia $(2, 0, 0)$, and $D_0(3) - D_0(2)$ has inertia $(1, 0, 1)$ (rank-one PSD). On an independent six-node toy, $d_0 = 3, 4, 5$ gives $\text{ind}_-(D_0) = 2, 1, 0$. A mass-imbalanced five-atom dual with the same two holes has inertia $(0, 2, 0)$ and $\det D_0 = 113/2880000 > 0$ (negative definite already in 2×2).*

Thus Gantmacher–Krein / variation-diminishing of the Cauchy– π residual does not forbid two negative directions, even in the smallest cases. The interpolating-depth residual is the *wrong depth* for the overload bound: on MAIN, $n_C(|Y|) = 43 \gg 1$.

3 The depth curve

Proposition 2 (source vs. adversaries). *On flagship MAIN,*

$$n_C(1) = 104, \quad n_C(|Y|) = 43, \quad n_C(181) = 1, \quad \min\{n : n_C(n) \leq 1\} = 178.$$

The last six values are $(2, 2, 1, 1, 1, 1)$ for $n = 176, \dots, 181$: the $2 \rightarrow 1$ drop is at $N_w - 6$, then three frozen ranks at 1. Λ -permute on the frozen mask: $n_C(|Y|) = 54$, $n_C(181) = 20$, never hits 1 (still dropping at the last step). Gaussian jitter ends at 19; position-scramble at 21 (different hole set $|Y| = 68$). A two-period world has $N_w - 3 = 8 < |Y| = 11$ and ends at $n_C = 4$ (working depth before interpolating depth).

The source and the adversaries obey the same rank-one algebra; they differ by *save rate* (how many of the $N_w - 4$ updates push an eigenvalue across 1). MAIN spends 103 of 180 updates on a crossing and finishes at 1; permute spends 83 and finishes at 20.

False depth: at $n = N_w - 1$ MAIN has $n_C = 0$ (the leftover square is liftable by extra OP). The cutoff $N_w - 3$ is the object, not a saturated identity.

4 Two-case census

Proposition 3 (selected two-case). *On the 42 admissible MAIN windows, $n_C(N_w - 3) \in \{0, 1\}$ on 42/42 (28 one-defect, 14 vacuous). The $2 \rightarrow 1$ locking rank (when it exists) sits at most 6 steps before $N_w - 3$; the $1 \rightarrow 0$ lift on vacuous windows sits at most 3 steps before $N_w - 3$. Living χ_3 37/37 and χ_4 41/41 keep $n_C \in \{0, 1\}$; dead χ_3 5/5 and χ_4 -20 likewise (death remains the edge, r401).*

No margin correlations are used.

5 What is a theorem, what is open

The nested CD chain, its rank-one Loewner updates, the interlacing bound on n_C , and the r407 dictionary $n_C(N_w - 3) = \text{ind}_-(A_0)$ are theorems. That a totally positive Cauchy– π residual has $\text{ind}_- \leq 1$ is closed (false). The selected save-rate $n_C(N_w - 3) \leq 1$ is a 42/42 census, not a source-pure identity: the remaining question is why the von Mangoldt dual OP sequence on Y spends enough updates on sub-threshold modes, while a permutation of the same masses does not. P1 stays $\lambda_2(C) \geq 1$ at degree $N_w - 3$. No RH claim.